# Anton Matiash

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## Professional Summary

Mechanical Engineering undergrad (Top 1% cohort) with a strong background in experimental physics (IYPT Gold Medalist). Focused on sim-to-real transfer and reinforcement learning for legged robotics. Passionate about bridging hardware and software through dynamic simulation, control theory, and modern machine learning techniques. Seeking a research internship to tackle complex challenges in autonomous robotics.

## Education

**BEng in Mechanical Engineering** *Sep 2024 – May 2027*
*Swansea University* *Swansea, UK*
First Year Average: 89/100(Top 1% cohort)
Most important first year modules: Applied Mechanics(95%), Data Analysis and Simulation(98%), Engineering Design 1(95 %), Thermofluids 1(81%), Engineering Mathematics(91%), Manufacturing Technology 1(89%)
Second Year Modules so far: Thermofluids 2(89%), Statistics(82%), Engineering Practice 1(92%)

**Modern Robotics specialisation (MR)** *June 2025 – Oct 2025*
*Northwestern University/Coursera* *Online*
Completed all 6 courses in specialisation: Foundations of Robot Motion, Robot Kinematics, Dynamics, Motion Planning and Control, Manipulation and Wheeled Mobile Robots, Capstone Project.

**High School Diploma** *Sep 2017 – June 2024*
*Kharkiv Physics-Mathematics Science Lyceum №27* *Kharkiv, Ukraine*
Last year Average: 11.1 out of 12

## Honors & Awards

**International Physicists' Tournament 2025** *Kielce, Poland*
Achieved 4th place among 17 countries worldwide as part of Team Ukraine, was solving mechanics problem about staircase motion of "Slinky" with experiments, simulations and theory over the year period and presented solution at the tournament achieving highest score for the report among my team.

**International Young Physicists' Tournament 2024** *Budapest, Hungary*
Achieved Gold Medal as part of Team Ukraine, competed among 38 countries worldwide, solved complex mechanics/fluids dynamics problem about soap film motion on complex shapes over the year period and presented it to the audience at the tournament achieving highest score for the report among my team.

**All-Ukrainian Student Physics Olympiad 2023/2024** *Ivano-Frankivsk, Ukraine*
Achieved 3rd place Diploma on the national stage, solved 4 theoretical and 2 experimental problems over the period of 6 hours, 2 points were missing to qualify for IPhO (which is the international stage of the Olympiad).

## Selected Projects

**Quadruped Robot design and Reinforcement Learning (RL) Project** *Oct 2025 – Present*
*ROS, Robotics, Engineering, Fusion360, C++, MuJoCo/IsaacSim, Python*

- Designing and building a custom 12-DOF quadruped robot utilising BLDC motors and LIDAR for advanced control and navigation via ROS. Currently designed and tested rotational actuator with 7:1 gear ratio.
- Developing a comprehensive Sim-to-Real pipeline by constructing accurate physical models in MuJoCo and Isaac Sim to facilitate safe policy training.
- Implementing Reinforcement Learning algorithms in Python to train robust locomotion policies for uneven terrain, serving as the core framework for my upcoming independent thesis research.

**Capstone Project (part of the MR specialisation)** *Jun 2025 – Oct 2025*
*MATLAB, Robotics, Computer Science* [Certificate Link][Demonstration Video]

- Engineered a complete kinematics and control software stack for a Kuka mobile manipulator in MATLAB, successfully executing automated pick-and-place tasks in CoppeliaSim.
- Developed a custom trajectory generator and PID controller using MATLAB to accurately manage joint limits and achieve precise end-effector positioning from randomized starting configurations.

**IMechE Foundation Design Challenge** *Jan 2025 – June 2025*
*Fusion360, Engineering, Physics(Electronics, Mechanics), Powerpoint* [Project poster]

- Designed and manufactured a car that could move forward to a point, stop, wait 15 seconds, and return to the start point, with purely analog control.
- Involved designing mechanical components, and custom electrical circuit (Complete information in poster).

**Other Projects**
**Mafia Bot** *TypeScript, Node.js, Discord.js, TypeORM* [GitHub], **Strategy Game** *C#, Godot Engine,* [GitHub]

## Technical Skills

**Engineering:** Fusion 360 (CAD & CAM), Wolfram Mathematica, CoppeliaSim, Grantu Edu-Pack, MATLAB Simulink, MuJoCo

**Programming Languages:** MATLAB, Wolfram Lang, C++, C#, TypeScript, Python, JS/HTML/CSS, SQL

**Frameworks & Libraries:** ROS, Pytorch, TypeORM, Svelte, Node.js, Electron.js, Godot

**DevOps:** Git, Docker, GitHub

**Languages:** English (C1), Russian (Native), Ukrainian (Native)

**Other:** LaTeX, Markdown, Microsoft Office

## Personal Interests

**Robotics & Technologies:** Passionate about reading robotics/technology news, watching videos and forums about.

**Politics & Economics & History:** History and current events of the world fascinate me, so I am keen on watching and analysing them.